Colby Cadden

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Education

University of Calgary

B.Sc. Mechanical Engineering / B.Comm. Finance; GPA: 4.0/4.0 — *Calgary, Alberta*

- **Schulich Leader Scholarship** — $120,000 national STEM scholarship recognizing academic excellence, leadership, and entrepreneurship.
- **President's Admission Scholarship** — awarded for an admission average of 95% or higher.

Salmon Arm Secondary

Dual Dogwood Diploma — *Salmon Arm, British Columbia*

- BC District/Authority Award ; BC Achievement Scholarship ; Governor General's Academic Medal

Engineering Projects

Sensor-Activated Robotic Arm | *Fusion 360, VS Code, ESP32, Arduino Nano, C++* **2025 – 2026**

- Designed and iterated a sensor-activated desktop robotic arm using ESP32 and Arduino Nano microcontrollers, ToF sensors, servo-driven joints, and 3D-printed PLA components.

Remote-Controlled Rescue Vessel | *Raspberry Pi Pico, Fusion 360, VS Code, 3D Printed Iteration* **2026**

- Led mechanical design and fabrication of an RC rescue vessel that navigated obstacles while carrying a payload

Hackathons | *Python, pandas, GNSS Data Processing, Anthropic API* **2026**

- **1st Place, G-Hacks 2026:** Processed RTK/PPP satellite data in Python to produce the most accurate positioning solution
- **2nd Place, IEEE Industry Hackathon 2026 (200+ Participants):** Built an autonomous LLM agent and data prioritization pipeline on real City of Calgary 311 data, improving safety coverage by 88% over a FIFO baseline

Leadership and Involvement

Mechanical Arm Team Member, *Schulich Space Rover Team* **2026 – Present**

- Iterating SolidWorks CAD for the rover arm, working with actuators, planetary gearboxes, and various manufacturing methods.

Student Representative, *University of Calgary* **2025 – 2026**

- Represented 200+ first-year engineering students by communicating feedback and collaborating with faculty to identify and implement solutions.

Athletic and Community Leadership, *Salmon Arm Secondary* **2021 – 2025**

- Captained varsity volleyball and ultimate frisbee varsity; assistant captain; varsity basketball.

Work Experience

Production Labourer, *Canoe Forest Products* **2026**

- Operated large production machinery and control panels across 4 stations, monitoring process sensors and alarms
- Cleared plug-ups and troubleshot equipment breakdowns to minimize downtime
- Followed lockout/tagout procedures when clearing and servicing equipment

Cone Picker, *Skimikin Seed Orchard* **2024 – 2025**

- Harvested pinecones using ladders and hand tools while meeting production targets and maintaining safe work practices.

General Labourer, *Mt. Ida Tree Nurseries Ltd. and Sorrento Nurseries Ltd.* **2023 – 2025**

- Packaged and handled nursery stock in fast-paced outdoor production environments.

Technical Skills and Certifications

Programming: Python, C++, MATLAB, Pandas, Numpy
Engineering Tools: Fusion 360, SolidWorks, Onshape, VS Code, Microsoft Excel
Hardware and Fabrication: ESP32, Arduino Nano, Raspberry Pi, Woodworking, Metalworking, Machining
Certifications: - DELF B2 French Language Proficiency
- Economic Fundamentals for Leadership
- Engineering Leadership and Development (Tier 2)